Electrostatics

Problem 2.1

(a) Twelve equal charges, q, are situated at the corners of a regular 12-sided polygon (for instance, one on each numeral of a clock face). What is the net force on a test charge Q at the center?

(b) Suppose on of your 12 qs is removed (the one at "6 o'clock"). What is the force on Q? Explain your reasoning carefully.

(c) Now 13 equal charges q, are placed at the corners of a regular 13-sided polygon. What is the force on a test charge Q at the center?

(d) If one of the 13 qs is removed, what is the force on Q? Explain your reasoning.

Solution:

(a) Due to the symmetry, the net force on a test charge at the center is zero. This is because each opposite charge q on the 12-sided polygon cancel each others force at the center. The force from the charge at 1 o'clock cancels the force from the charge at 7 o'clock, etc.

(b) If the charge at 6 o'clock is removed, the net force on a test charge at the center is

$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}$$

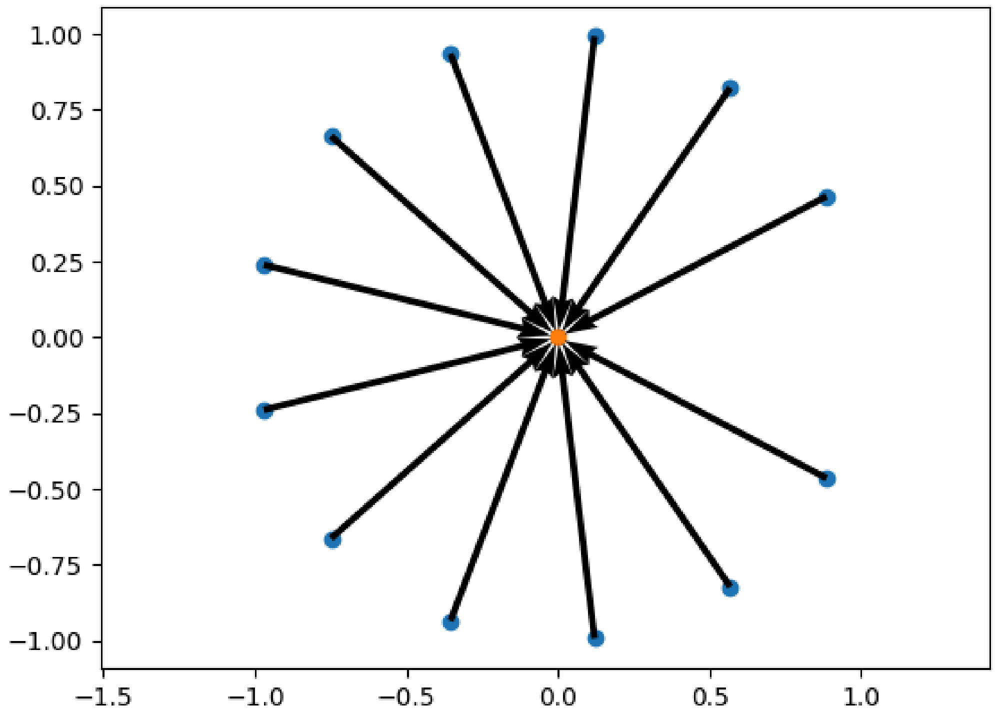
directed straight down (straight up) if the charge q and the test charge Q are both positive or both negative (if they are positive/negative or negative/positive respectively).

(c) Again, due to symmetry, the net force at the center is zero.

(d) The force on the center would again be

$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}$$

directed towards the removed charge (or opposite direction, if the charges q and the test charge Q are of opposite signs). This can either be explained by simply adding a charge of opposite sign at the removed charge, or by the same idea as in (b), from the following image



All vertical components of the forces cancel, giving the total force as the sum of the individual charges:

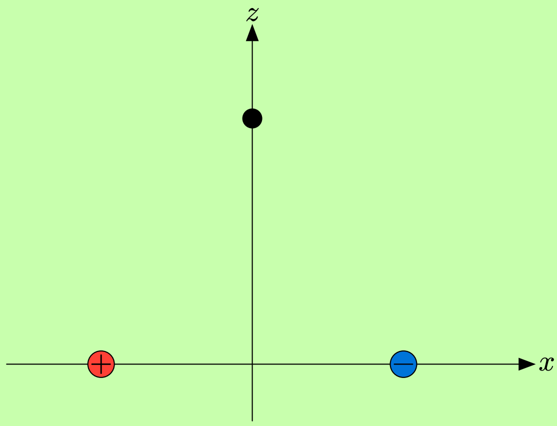
$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}\sum_{k=1}^{13}\cos\left(\frac{2\pi k}{13}\right) = \frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}\hat{\mathbf{r}} \text{ with } \hat{\mathbf{r}} \text{ pointing towards the right}$$

□

Problem 2.2

Find the electric field (magnitude and direction) a distance z above the mid-point between equal and opposite charges ($\pm q$), a distance d apart.

Solution:



The electric field from a point charge is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}$$

where $\mathscr{r}$ is the distance between the charge and the field point and $\hat{\boldsymbol{\mathscr{r}}}$ is the unit separation vector from the charge to the field point. The field at our measurement point will have no vertical (z-directed) component since the positive charge will create a field in the positive z-direction with the same magnitude that the negative charge will create a field in the negative z-direction. Therefore the total field will be directed towards $+x$ with a magnitude twice that of what the positive charge creates. The separation vector magnitude becomes

$$\mathscr{r} = \sqrt{z^2 + \left(\frac{d}{2}\right)^2}$$

and

$$\boldsymbol{\mathscr{r}} = \frac{d}{2}\hat{\mathbf{x}}$$
$$\downarrow$$
$$\hat{\boldsymbol{\mathscr{r}}} = \frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}} = \frac{d}{2\,\mathscr{r}}\hat{\mathbf{x}}$$

resulting in the total electric field from both charges

$$\boxed{\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{qd}{\sqrt{z^2 + \left(\frac{d}{2}\right)^2}^3}\hat{\mathbf{x}}}$$

$\square$

Problem 2.3

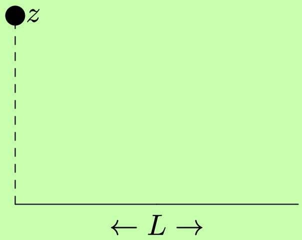
Find the electric field a distance z above one end of a straight line segment of length L that carries a uniform line charge λ. Check that your formula in consistent with what you would expect for the case $z \gg L$.

Solution:

The electric field from a continuous line charge distribution is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{\mathscr{r}^2} \, \mathrm{d}l'$$

The setup is shown in the figure below



The separation vector $\boldsymbol{\mathscr{r}}$ becomes

$$-x\hat{\mathbf{x}} + z\hat{\mathbf{z}}$$

which gives

$$\hat{\boldsymbol{\mathscr{r}}} = \frac{-x\hat{\mathbf{x}} + z\hat{\mathbf{z}}}{\sqrt{x^2 + z^2}}$$

This means our integral becomes, since λ is constant,

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{-x\hat{\mathbf{x}} + z\hat{\mathbf{z}}}{\sqrt{x^2 + z^2}^3} \, \mathrm{d}x$$

The following standard integrals will be used in the calculations:

$$\int \frac{1}{\sqrt{x^2 + a^2}^3} \, \mathrm{d}x = \frac{1}{a^2} \frac{x}{\sqrt{x^2 + a^2}}$$

$$\int \frac{x}{\sqrt{x^2 + a^2}} \, \mathrm{d}x = -\frac{1}{\sqrt{x^2 + a^2}}$$

Starting with the x-component gives

$$E_x = -\frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{x}{\sqrt{x^2 + z^2}^3} \, \mathrm{d}x = -\frac{\lambda}{4\pi\epsilon_0} \left[-\frac{1}{\sqrt{x^2 + z^2}} \right]_0^L =$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{z} - \frac{1}{\sqrt{L^2 + z^2}} \right)$$

And the z-component

$$E_z = \frac{\lambda}{4\pi\epsilon_0}\int_0^L \frac{z}{\sqrt{x^2+z^2}^3}\,\mathrm{d}x = \frac{\lambda z}{4\pi\epsilon_0}\left[\frac{1}{z^2}\frac{x}{\sqrt{x^2+z^2}}\right]_0^L =$$

$$= \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{\sqrt{L^2+z^2}}\right)$$

The electric field is then

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z} - \frac{1}{\sqrt{L^2+z^2}}\right)\hat{\mathbf{x}} + \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{\sqrt{L^2+z^2}}\right)\hat{\mathbf{z}}$$

In the case that $z \gg L$, we get

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z} - \frac{1}{z}\right)\hat{\mathbf{x}} + \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{z}\right)\hat{\mathbf{z}} = \frac{\lambda L}{4\pi\epsilon_0 z^2}\hat{\mathbf{z}}$$

which is of the same form as the field at a point z from a point charge of magnitude λL. □

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